University of Basel

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Introduction to Databases — Autumn 2025

Exercise 1

(Theory) **Hand-in: 08.10.2025 (ADAM, 23:59)**
(Practical) **Hand-in: 08.10.2025 (during exercise)**

Solving the Exercises: The exercises can be solved in small groups of a maximum of two people. Use the notations introduced in the lecture. The DMI plagiarism guidelines apply for this lecture.

Submission Information: Please upload all (Theory) deliverables BEFORE the deadline to ADAM as **a single PDF**. For (Practical) exercise check the **Hand-In** instructions. The solutions must contain your names. Solutions that are handed in too late cannot be considered.

Task (Practical) 1: Groups (1 Points)

This will be by far the easiest task. Form groups of two and let the tutors know to get your first point. Be careful with your choice: The groups are fixed for the duration of the course.

Hand-In: Register your group by sending a mail with your names and unibas addresses to one of the tutors. In return you will get credentials to connect to the database. Do not share your credentials outside your group.

Task (Practical) 2: Connect to the database (2 Points)

In this task you need the credentials obtained in the first task.

You will now connect to the database (running on PostgreSQL) and execute a few queries. The database is only reachable from inside the university VPN. Use a tool of your choice, e.g. the open-source community edition of dbeaver. Other options are JetBrains DataGrip, or psql if you prefer a command line tool. Use the credentials/details obtained in the previous task.

To verify that you are correctly connected, run the following SQL command:

```sql
SELECT version();
```

In exercise 2 we will learn how to connect to the database using Python.

Hand-In: Show the assistants/tutors that you can execute the version command. What database version is used?

Task (Practical) 3: Exploring the database (3 Points)

For this task it is required to complete the previous tasks first.

Now that you are successfully connected, have a look around. What data is stored inside this database?

Now execute the following query. Can you guess what it does?

```
SELECT COUNT(*)
  FROM games
 WHERE result = '1-0'
```

Hand-In: Explain the assistants/tutors what data is in the database. What does this query do?

Task (Practical) 4: Exploring the database II (1 Points)

Now execute the following query and plot the result with a program of your choice.

```
SELECT EXTRACT(MONTH FROM game_start AT TIME ZONE 'UTC'), COUNT(*)
  FROM games
GROUP BY EXTRACT(MONTH FROM game_start AT TIME ZONE 'UTC')
```

Hand-In: What does this query do? Show the plot to the assistants/tutors.

Task (Theory) 5: Questions (3 Points)

Answer the following questions and explain your answers:

- What are the advantages and disadvantages of using databases to manage data compared to using spreadsheets? *(2 advantages and 2 disadvantages as bullet points)*
- What is the link to the official documentation for the `version()` function used in the first task and to which category of functions does it belong?

Hand-In: Hand in your solutions via ADAM.